# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
## Department of Computer Science and Engineering

## ASSESSMENT TOOL 3– Output Prediction Questions

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| **Course Code:** | CSA0315 | **Course Title:** | Data Structures |
| **CO Assessed:** | CO1 | **Assessment:** | Assessment Tool 3 – Debugging and Optimization Questions |
| **Weightage:** | 10% of CIA | **Total Marks:** | 100 |
| **CO2 Statement:** | Basics, Data, Data object, Data structure, Abstract Data Types (ADT), realization of ADT in 'C', Primitive and non-primitive data structure, Linear and non-linear data structure Arrays&Recursion, Performance analysis and Measurement: Time and space analysis of algorithms, Average, best and worst-case analysis. | | |
| **Student Name:** | Nandha Krishna V | **Reg. No.:** | 192524277 |
| **Section / Batch:** | | **Date:** | |

**Code of Conduct**

*I ________________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: ________________________

**Instructions**

- This test contains 10 MCQ Based Questions. Total: 100 Marks.
- When a student answer using saveetha compiler, execute the code, Test the test cases , While evaluation the answer should contain following four requirements: Understanding of concepts, Connections between ideas, Logical structure, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.
- Electronic Gadgets are permitted in the class. Take it as test and submit in the saveetha lab compiler class.

## Annexure A – Output Prediction Questions

1.Write a C program that takes an array as input and returns it reversed.

2.Write a C program for palindrome

3.Write a C program  for Searching  an Element in a Sorted Array

4.Write a C program  for Count the Occurrences of an Element

5.Write a program for binary search

6.Write a C program  for Merge Two Arrays

7.Write a C program  for Find the Largest and Smallest Element in an Array

8.Write a C program  for Find the Second Largest Element

9.Write a C program  for Find the First and Last Occurrence of an Element

10.Write a C program  for Find Missing Element in an Array

## Rubrics for Calculation

| Criteria | Weight age | Level 4 – Exemplary | Level 3 – Proficient | Level 2 – Developing | Level 1 – Beginning |
|---|---|---|---|---|---|
| Understanding of Problem Context | 20 | Clearly understands the problem and accurately identifies all requirements | Good understanding with minor gaps | Basic understanding; some requirements unclear | Poor understanding or misinterpretatio n of the problem |
| Application of Concepts | 30 | Accurately applies appropriate concepts to solve complex problems | Applies relevant concepts correctly with minor errors | Applies basic concepts with limited accuracy | Fails to apply appropriate concepts |
| Problem-Solving Approach | 20 | Logical, well-structured, and efficient approach | Mostly logical approach with minor gaps | Basic approach with some inconsistencies | Illogical or unclear approach |
| Accuracy of Solution | 20 | Completely correct solution with proper justification | Mostly correct with minor errors | Partially correct solution | Incorrect or incomplete solution |
| Clarity | 10 | Clear, well-organized, and properly explained solution | Understandable with minor clarity issues | Limited clarity and explanation | Poorly presented and difficult to understand |

| Faculty In-charge | Course Coordinator | Program Director |
|---|---|---|
| | | |
| Signature: _______________ | Signature: _______________ | Signature: _______________ |
| Date: ___________________ | Date: ___________________ | Date: ___________________ |

```c
#include <stdio.h>

int main() {
    int n, i, key;
    int arr[100];
    int low, high, mid;
    int found = 0;

    printf("Enter the size of array: ");
    scanf("%d", &n);

    printf("Enter %d sorted elements:\n", n);
    for (i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    printf("Enter the element to search: ");
    scanf("%d", &key);


    low = 0;
    high = n - 1;

    while (low <= high) {
        mid = (low + high) / 2;

        if (arr[mid] == key) {
            found = 1;
            break;
        }
        else if (key < arr[mid]) {
            high = mid - 1;
        }
        else {
            low = mid + 1;
        }
    }

    if (found) {
        printf("\nElement %d found at index %d\n", key, mid);
        printf("Position: %d\n", mid + 1);
    } else {
        printf("\nElement %d not found in the array.\n", key);
    }

    return 0;
}
```

```c
#include <stdio.h>

int main() {
    int n, i, key;
    int arr[100];
    int first = -1, last = -1;

    printf("Enter the size of array: ");
    scanf("%d", &n);

    printf("Enter %d elements:\n", n);
    for (i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    printf("Enter the element to search: ");
    scanf("%d", &key);


    for (i = 0; i < n; i++) {
        if (arr[i] == key) {
            if (first == -1)
                first = i;
            last = i;
        }
    }

    if (first != -1) {
        printf("\nElement %d found.\n", key);
        printf("First occurrence: Index %d (Position %d)\n", first, first + 1);
        printf("Last occurrence : Index %d (Position %d)\n", last, last + 1);
    } else {
        printf("\nElement %d not found.\n", key);
    }

    return 0;
}
```

**INPUT**

```
8
10 20 30 20 40 20 50 60
20
```

```c
#include <stdio.h>

int main() {
    int n1, n2, i;
    int arr1[100], arr2[100], merge[200];


    printf("Enter size of first array: ");
    scanf("%d", &n1);

    printf("Enter %d elements of first array:\n", n1);
    for (i = 0; i < n1; i++) {
        scanf("%d", &arr1[i]);
    }


    printf("Enter size of second array: ");
    scanf("%d", &n2);

    printf("Enter %d elements of second array:\n", n2);
    for (i = 0; i < n2; i++) {
        scanf("%d", &arr2[i]);
    }


    for (i = 0; i < n1; i++) {
        merge[i] = arr1[i];
    }

    for (i = 0; i < n2; i++) {
        merge[n1 + i] = arr2[i];
    }


    printf("\nMerged Array:\n");
    for (i = 0; i < n1 + n2; i++) {
        printf("%d ", merge[i]);
    }

    printf("\n");

    return 0;
}
```

**INPUT**

```
3
10 20 30
4
40 50 60 70
```

## CO2 - AT2 - Code Debugging

CLOSED ✓ Completed

Closed

50/50 answered

| YOUR MARKS | CLASS AVERAGE |
| --- | --- |
| **80** / 100 | **69.1** / 100 |

View Rubric